

Mathematics A

Unit 1 • Chapter OL-T02 • Expertise Q16+ Classified

Cross-session • chapter-organised candidate

3 classified items • 7 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

Every question is followed by its worked answer/reference

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Contents

Unit 1 • Expertise • Unit 1 • Chapter OL-T02 • Expertise Q16+ • 3 items • 7 marks

Chapter	Items	Marks	Starts
01. OL-T02 Set Notation & Venn Diagrams	3	7	4

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Set Notation & Venn Diagrams

Unit 1 • Expertise • 7 marks • 3 item(s)

January 2021 | 4MA1-2H Q16 | 3 marks

Complete or solve a practical three-set survey

January 2021 | 4MA1-2H Q16 | 3 marks

Complete or solve a practical three-set survey

June 2021 | 4MA1-1H Q19 | 1 marks

Read and combine three-set regions

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

16 Some students were asked the following question.

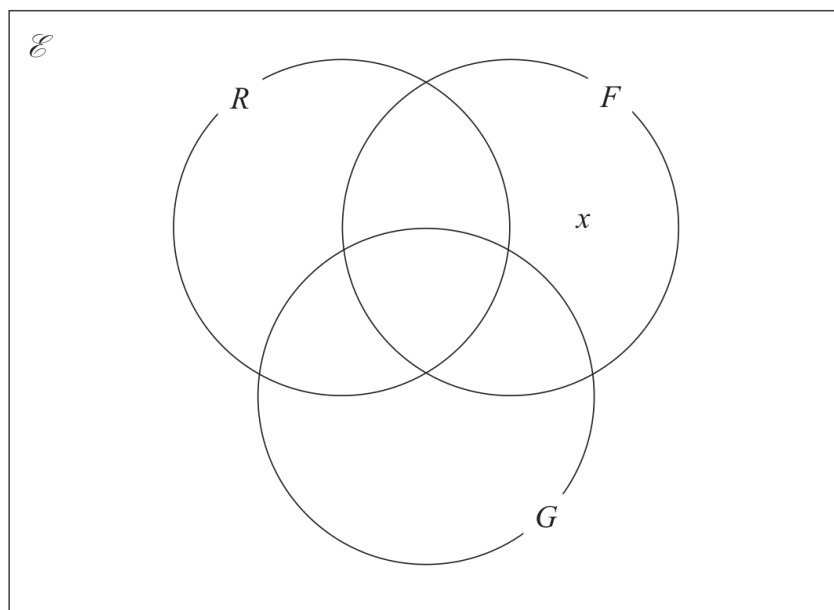
“Which of the subjects Russian (R), French (F) and German (G) do you study?”

Of these students

- 4 study all three of Russian, French and German
- 10 study Russian and French
- 13 study French and German
- 6 study Russian and German
- 24 study German
- 11 study none of the three subjects
- the number who study Russian only is twice the number who study French only.

Let x be the number of students who study French only.

(a) Show all this information on the Venn diagram, giving the number of students in each appropriate subset, in terms of x where necessary.



(3)

Given that the number of students who were asked the question was 80

(b) work out the number of these students that study Russian.

(3)

Worked solution

January 2021 | Question 16 | 3 marks

Family: Resolve a three-set Venn model • Variant: Complete or solve a practical three-set survey

Method / working

1. Place 4 in the triple intersection, $10-4=6$ in R intersection F only, $13-4=9$ in F intersection G only, and $6-4=2$ in R intersection G only.

2. Let French-only= x and Russian-only= $2x$; German-only= $24-4-9-2=9$.

3. Place 11 outside and label every region on the supplied Venn diagram.

Final answer: Completed Venn diagram with x , $2x$, 6, 2, 4, 9, 9, 11

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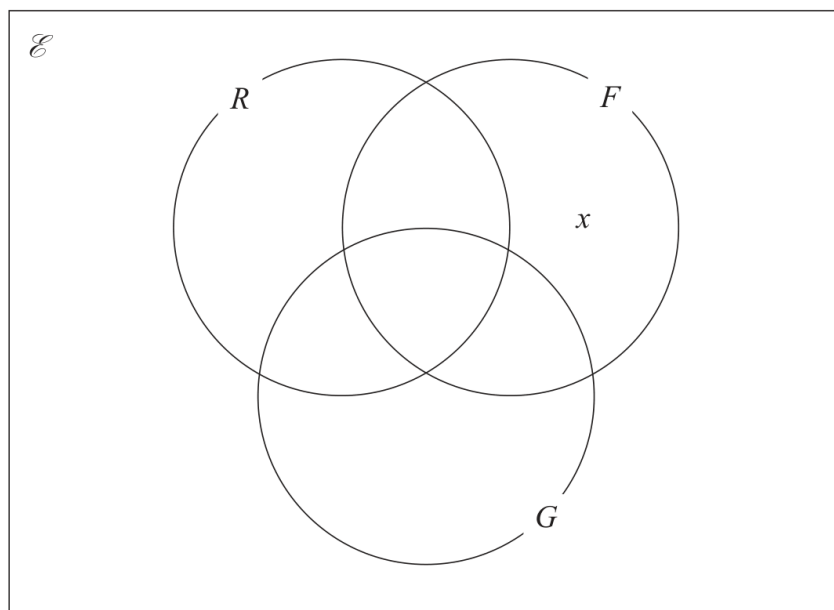
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Worked solution

January 2021 | Question 16 | 3 marks

Family: Resolve a three-set Venn model • Variant: Complete or solve a practical three-set survey

Method / working

1. Use the total: $2x+6+x+2+4+9+9+11=80$.

2. Solve $3x+41=80$ to get $x=13$.

3. Russian total $=2x+6+2+4=38$.

Final answer: 38 students

(iii) $n([A \cap B] \cup C)$

(1)

Worked solution

June 2021 | Question 19 | 1 marks

Family: Resolve a three-set Venn model • Variant: Read and combine three-set regions

Method / working

1. Count $(A \cap B) \cup C$ from the labelled regions: 28.

Final answer: 28